

Senior Software QA Engineer – Practical Exercise

Part 1: QA Strategy & Release Readiness

Assumptions and clarifications

- Ages 15–24 are inclusive. Phone numbers follow the supported country format; shared phones may exist, so phone number alone should not determine whether two participants are duplicates.
- “Immediate SMS” needs a measurable delivery target agreed with the Product Owner and SMS provider. I propose, as a starting point, 95% of messages delivered within 30 seconds and 99% within 2 minutes. Provider acceptance and receipt on the participant’s phone should be measured separately.
- USSD requires mobile network coverage even when internet access is unavailable.
- Attendance behavior is not specified in the acceptance criteria. I would confirm its scope before declaring the broader feature complete.

1. Definition of Done

Before production release:

- ☐ **Registration validation:** Web and USSD registration collect phone number, full name, age, and district. Ages 15 and 24 pass; 14 and 25 fail. Missing or invalid inputs produce clear errors, with validation enforced on the server.
- ☐ **Offline durability:** Web registrations persist locally during connection loss and survive a page reload or browser restart. Mentors can distinguish pending, synced, and failed registrations.
- ☐ **Reliable synchronization:** Pending registrations sync automatically when connectivity returns. Interrupted requests, retries, and repeated submissions create one record per registration. Queue entries are removed only after confirmed server success.
- ☐ **USSD/SMS correctness:** Successful USSD registration creates a unique six-digit Participant ID and sends confirmation within the agreed delivery target. IDs retain leading zeros, and uniqueness is enforced centrally. Failed delivery can be retried without creating another participant.
- ☐ **Field usability:** Registration works on representative low-end phones, small screens, supported browsers, and weak connections. Required controls remain visible and usable.
- ☐ **Performance under field conditions:** Registration and sync are tested under throttled 2G/3G, high latency, and packet loss, including a bulk sync of a large queued backlog.
- ☐ **Data protection:** Only authorized mentors can access participant data. Locally stored data is protected appropriately, retention and logout behavior are defined, and logs avoid exposing personal details.
- ☐ **Verification:** Acceptance tests and regression tests pass. No unresolved release-blocking defects remain; other known issues have documented impact, ownership, and Product Owner acceptance.
- ☐ **Field pilot:** A pilot with real field mentors on their own devices and networks is completed, and its findings are triaged before full rollout.
- ☐ **Operational readiness:** Sync failures, duplicate creation, and SMS delivery latency are monitored. Recovery and rollback procedures are documented, and QA and the Product Owner review release evidence.

2. Two Key Failure Risks

Risk	Impact	Mitigation and verification
The server saves a registration, but connectivity drops before the browser receives confirmation. The browser retries when online.	Duplicate participants can distort attendance and reporting; deleting the queued entry prematurely could instead lose a registration.	Assign a stable idempotency key when the registration is first queued and reuse it on every retry. Enforce it server-side. Test connection loss before submission, after server commit, and during repeated reconnections; verify exactly one record and correct queue status.
USSD registration succeeds, but the SMS is delayed or fails. The participant repeats registration because they received no confirmation.	Participants may receive conflicting IDs, duplicate records, or no usable confirmation.	Show the assigned ID in the successful USSD response where possible. Persist a notification job with the registration, track delivery status, and retry failed messages independently. Test provider timeouts, delayed delivery, and repeated sessions; verify notification retries preserve the original participant and ID.

3. Bug Severity and Priority

Severity measures impact; priority determines how urgently to fix the defect. I would use this classification:

- **S1 Critical:** System-wide failure, unrecoverable data loss, or severe security impact.
- **S2 Major:** Core functionality is blocked or data integrity is compromised.
- **S3 Moderate:** Functionality is degraded, with limited impact or a viable workaround.
- **S4 Minor:** Cosmetic or low-impact usability issue.

Priorities: P1 release blocker, fix before release; P2 fix before release or ship with a Product Owner–approved mitigation; P3 schedule for the next planned release or backlog.

Bug	Severity	Priority	Reasoning
A: District dropdown renders off-screen on low-end mobile browsers	S2 Major (assuming users cannot select a district)	P1	District is required, so affected mentors cannot complete web registration. Low-end devices are relevant to the intended field users. If scrolling reliably exposes the control, I would reassess it as S3/P2 after validating that workaround.
B: Reconnection creates duplicate participant records	S2 Major	P1 – highest fix order	This compromises data integrity during an expected rural-connectivity scenario and can affect attendance and reporting. Block release until retries are idempotent, regression tests pass, and any existing duplicates have been assessed.
C: Confirmation SMS takes up to four minutes	S3 Moderate (assuming registration succeeds and the SMS eventually arrives)	P2	The core record remains intact, so this is less urgent than A and B, but delayed confirmation violates the stated acceptance criterion and may cause repeated registration attempts. Investigate application queues and provider delivery separately. Release requires meeting the agreed delivery target or explicit Product Owner acceptance of a mitigation, such as showing the ID in the USSD response.

Release recommendation

Hold release until Bug B is fixed, Bug A is resolved on supported field devices, and Bug C meets the agreed delivery target or receives documented acceptance of a mitigation or revised requirement. Fix order: B first because it corrupts data, then A because it blocks registration, then C.